# Lakshya Sharma

Jaipur, Rajasthan | 9116775258 | lakshyasharma593@gmail.com | LinkedIn | GitHub

## SUMMARY

AI/ML Engineer with hands-on experience building and evaluating deep learning models in NLP and computer vision, and deploying GenAI systems via REST APIs. Skilled in Python, PyTorch, TensorFlow, and Scikit-learn, with experience in LangChain, Hugging Face, and vector databases for RAG workflows. Strong foundation in neural networks (CNN, LSTM), model evaluation (F1, LLM-as-judge), MLOps (Docker), SQL, and AWS — with a collaborative, problem-solving mindset.

## TECHNICAL SKILLS

- **Languages:** Python, Java, SQL
- **Machine Learning & Deep Learning:** Classification, Regression, Clustering, Model Evaluation & Benchmarking, Neural Network Training/Evaluation, CNNs, RNNs, LSTM, PyTorch, TensorFlow/Keras, scikit-learn, XGBoost
- **Computer Vision:** OpenCV, YOLOv8, CNN-based Image Classification
- **NLP:** Text Preprocessing, Embeddings (Sentence-Transformers), Semantic Search
- **Generative & Agentic AI:** Prompt Engineering, LLM Evaluation, RAG, Agentic Workflows, Multi-Agent Systems, Multimodal AI, LangChain, LangGraph, Hugging Face Transformers, LLM API Integration
- **Backend & APIs:** FastAPI, Flask, Spring Boot, Node.js/Express, REST API Design, ONNX, Inference Optimization, Systematic Experiment Logging
- **Data & Storage:** SQL/NoSQL, MongoDB, ChromaDB (Vector Search), Data Preprocessing & Feature Engineering
- **Cloud & Tools:** AWS, GCP (CDAC-certified), Docker, Git/GitHub, Linux (Ubuntu), Postman

## EDUCATION

**Vellore Institute of Technology (VIT)** — Chennai, Tamil Nadu
*Master of Computer Applications (MCA) - CGPA: 8.19* — 2024–2026

**Aryan College** — Ajmer, Rajasthan
*Bachelor of Computer Applications (BCA) - 67.57%* — 2021–2024

## PROJECTS

**Federated RAG Incident Report System** *Python, PyTorch, LangChain, FastAPI, ChromaDB, React*

- Architected an end-to-end privacy-preserving incident reporting system combining Federated Learning and RAG, generating structured 4-section reports in **under 60 seconds** across **13 federated clients** without centralising raw data.
- Implemented a multimodal AI pipeline integrating YOLOv8 evidence extraction and Gemini 2.0 Flash LLM generation; evaluated **retrieval configurations** to achieve **90% contextual relevance** on policy citation retrieval using a structured evaluation framework.
- Developed a hybrid RAG retrieval stack (BM25 + Cross-Encoder re-ranking + ChromaDB) and conducted structured prompt experiments, logging results to select the highest-coherence output format.
- Containerised the full FastAPI backend with Docker and exposed the multimodal pipeline as REST endpoints, enabling reproducible deployment across environments.

**Customer Churn Prediction & MLOps Platform** *Python, Scikit-learn, Pandas, MLflow, FastAPI, Docker*

- Developed an end-to-end **customer churn prediction pipeline** using Python, Pandas and Scikit-learn, performing EDA, data preprocessing and feature engineering on customer behavioral and transactional data.
- Benchmarked **Logistic Regression, Random Forest and Gradient Boosting** models using precision, recall, F1-score and ROC-AUC, selecting the best-performing model for production inference.
- Implemented **MLflow-based experiment tracking and model versioning**, enabling reproducible comparison of training runs and controlled promotion of the best model.
- Built a **FastAPI REST service** for real-time and batch churn prediction, containerised the application using Docker and added automated API/model validation using PyTest.

**ResearchMind - Multi-Agent AI Research Pipeline** *Python, LangChain, LangGraph, Flask*

- Developed a **4-stage multi-agent research system** orchestrating specialised Search, Reader, Writer, and Critic agents to automate end-to-end research report generation from a single topic input.
- Created a **tool-using Search Agent** to autonomously retrieve recent, reliable web information, and a **Reader Agent** that dynamically selects the most relevant source and extracts detailed content via automated scraping.
- Orchestrated an **LLM-powered Writer chain** to synthesise combined search and scraped content into a structured research report, paired with a **Critic chain** that evaluates output against defined quality criteria, forming an automated LLM evaluation framework.
- Structured a modular workflow architecture with **shared state management** across agents, enabling seamless data flow from raw search results to a critiqued final report.

## CERTIFICATIONS

- Generative AI with Large Language Models – Coursera
- Python for Data Science, AI & Development – Coursera
- Agentic AI Foundations Associate – Oracle
- Cloud Architecture – CDAC Chennai (AWS/GCP fundamentals)

## EXTRACURRICULAR ACTIVITIES

- Management Team Member, E-Cell Club, VIT Chennai — coordinated large-scale events, demonstrating strong communication and team collaboration skills.
- Volunteer, Red Ribbon Club — contributed to awareness drives and campus social initiatives.